Test Normal Cases

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected routine-testing record.

Prepare R-case expectations, collect both methods' traces, audit Q0 and judge the independent equipment record.

Keep one habit: Keep expected outputs, actual reasons, pass judgments, versions and failures together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Build a Paper Prototype

Choose Inputs and Outputs

Verify Important Outputs

W1 · Prepare routine expectations before running

> Use the standalone catalogue and define the full-pass criterion.

Your evidence record

- Standalone activity-test reference: B1 Bookmark drawing (10 min, quiet Yes), B2 Story circle (8 min, quiet No), B3 Pattern puzzle (12 min, quiet Yes), B4 Tower challenge (15 min, quiet No), in that order. Valid minutes are whole 0–30; Quiet required is Yes/No. Require time fit AND (quiet not required OR activity quiet), choose first fit, otherwise no match; invalid input needs clarification. P1 validates, resets at B1 each request, advances after failure and stops at the first fit or end; the manual baseline marks every row before choosing the first fit. R1= (11,Yes), R2=(9,No), R3=(6,No), R4=(9,Yes). A full pass requires the correct output and supported explanation.

- Create the test record for R1 11 Yes, R2 9 No, R3 6 No and R4 9 Yes. Before running anything, write each expected **output**, explanation and purpose using catalogue B version 1.

  Include blank places for observed **output**, observed reason, pass or fail and trace reference. State the version labels and **pass criterion**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Run the same requests through both methods

> Record actual traces without inventing observations.

Your evidence record

- Operate your saved P1 cards and the manual **baseline** on all four R cases. Record **actual result**s and traces on your sheet.

  Summarise full passes over all attempted cases for each method, name any failures and explain why a tie would not prove the prototype improves correctness. Do not invent an observed run.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit the supplied Q0 comparison

> Keep the failed case and explain the broken quiet condition.

Your evidence record

- Supplied fictional Q version 0 record for W3: R1→B1 with correct time/quiet reason; R2→B2 with correct time/unrestricted-quiet reason; R3→no match because all durations exceed 6; R4→B2 because 8 fits 9, ignoring required quiet. The fictional manual baseline had the correct output and supported reason on all four cases. These records are not observations of your own P1 runs.

- Use the fictional Q version 0 record. Calculate its full-pass fraction and the **baseline**'s fraction. Explain why R4 fails and why dropping it to report 3 of 3 misrepresents the original four-case set.

  State what **evidence** you would inspect before changing Q.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Explain reachable outputs and coverage limits

> Distinguish visiting a row from selecting it as the first fit.

Your evidence record

- Explain why catalogue B cannot produce B3 or B4 as its first **fitting output** even if you try many ordinary requests. Distinguish checking a row during a failed search from successfully selecting it.

  Write a limited conclusion for a P1 run that passed all four R cases and name a useful next test category.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit a new test record

> Keep one first attempt before hints or feedback.

Read and explain independently

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”.

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a new test record

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Create and run the sorter's normal-case test plan with unique IDs and version label.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Run a fair comparison for Testing routine cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Testing routine cases. Proposed explanation: “A correct label is always a full pass”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Four routine cases are chosen before testing. Three are correct.

What can be reported?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 33 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”.

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional pencil-pouch finder has been run on four ordinary requests.

Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9.

Requests 2,4,6,8; recorded **output**s P1,P2,P1,none. These are invented records, not class results.

AI ACADEMY / AI-1/33

Expected versus recorded routine outputs

	Request	Expected	Recorded	Result
Test 1	2	P1	P1	pass
Test 2	4	P2	P2	pass
Test 3	6	P2	P1	fail
Test 4	8	none	none	pass

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/33

Test Normal Cases

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Write expectations

For 2,4,6,8 expect P1,P2,P2,none.

2. Compare outputs

Runs at 2,4 and 8 agree; run at 6 disagrees.

3. Trace failure

P1 holds only 3, so it cannot satisfy request 6.

4. Report

Three of four routine cases pass; retain the failure.

CHANGE ONE THING

Prototype reports none for request 6 instead of P1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is none correct for 8 but incorrect for 6? Use both capacity cards.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional pencil-pouch finder has been run on four ordinary requests.
- Which labelled step matches this detail? Three of four routine cases pass; retain the failure.
- 1. Trace failure
- 2. Write expectations
- 3. Compare **output**s
- 4. Report
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report Three of four routine cases pass; retain the failure. An ordinary case is not automatically correct.

Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires.

Changed-case question

- Changed-case practice: Prototype reports none for request 6 instead of P1. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. That run still fails.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That run still fails. P2 can hold 6, so none is also wrong under the first-fit rule.

Repair and save your work

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the repair?.

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

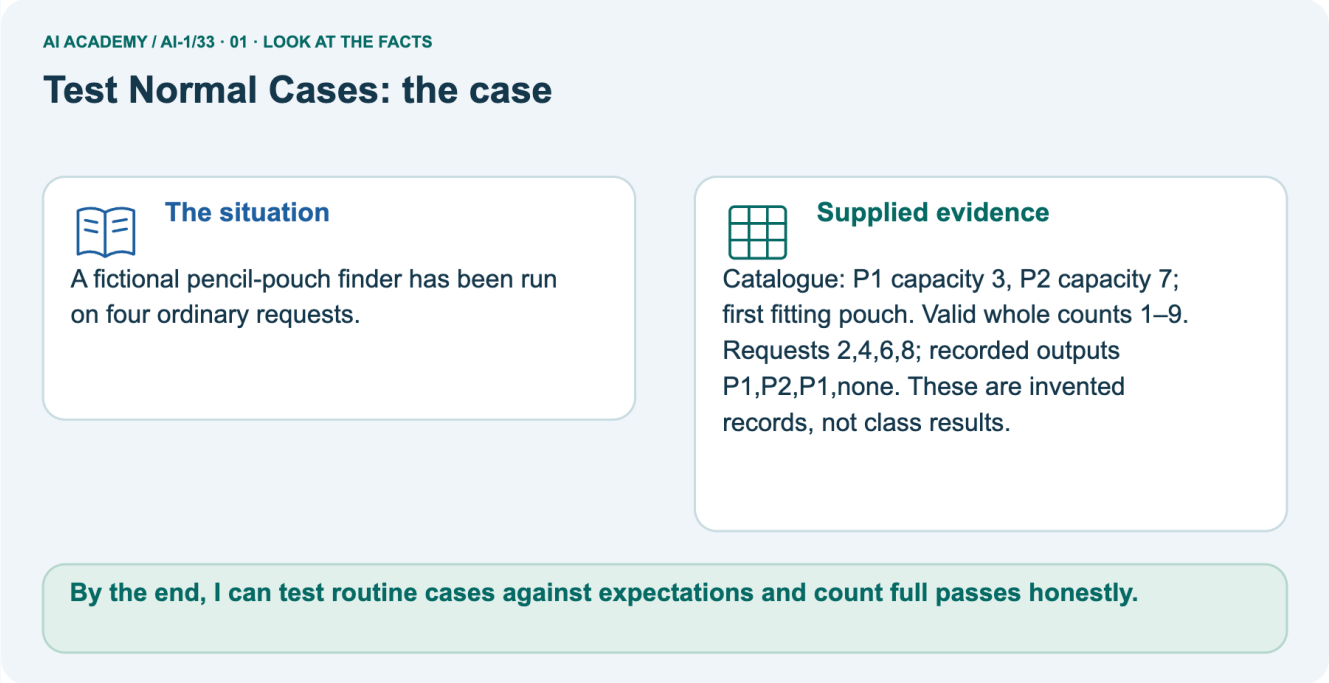
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Test Normal Cases: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

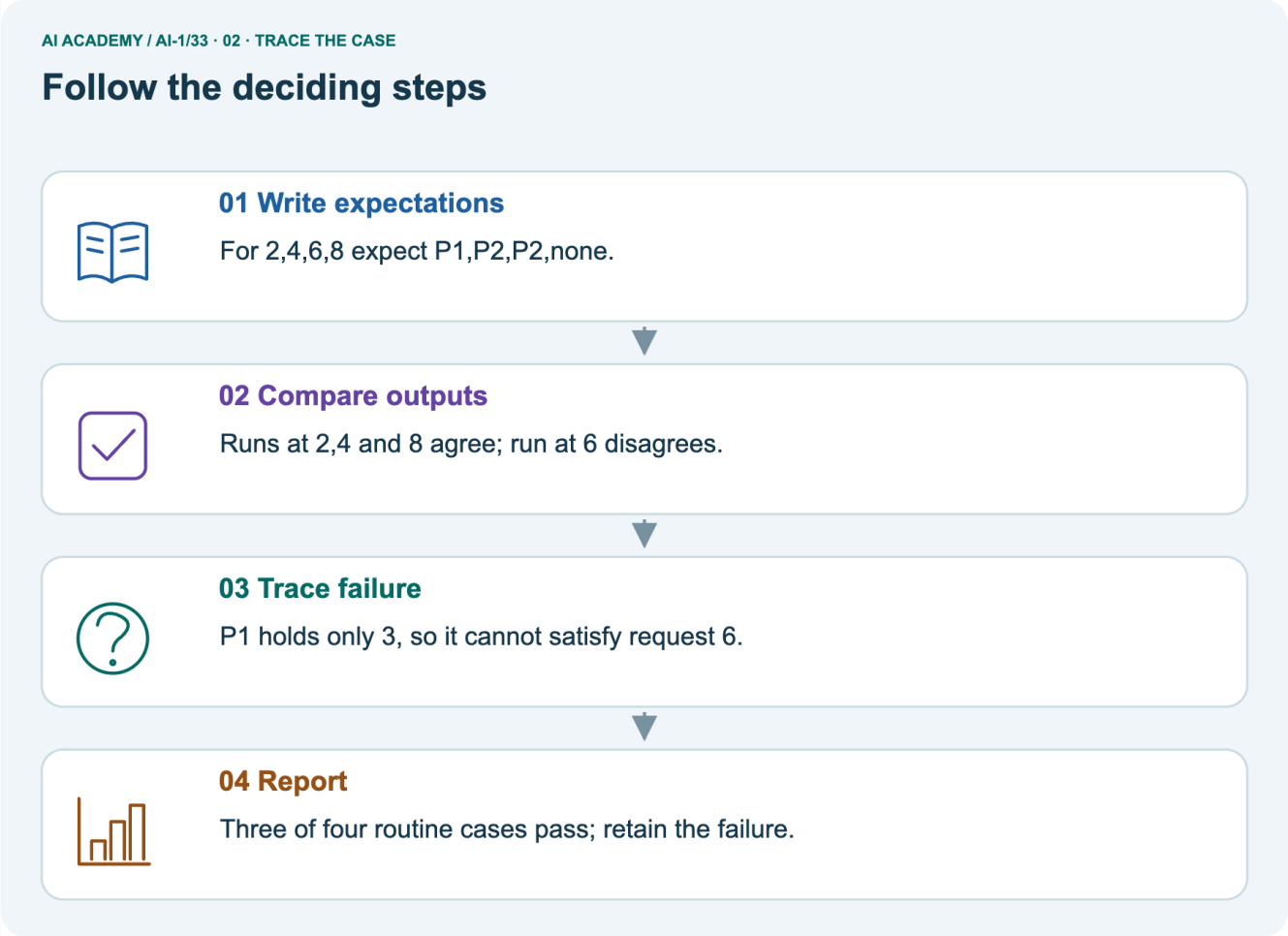
A fictional pencil-pouch finder has been run on four ordinary requests. Supplied facts: Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9. Requests 2,4,6,8; recorded **output**s P1,P2,P1,none.

These are invented records, not class results.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



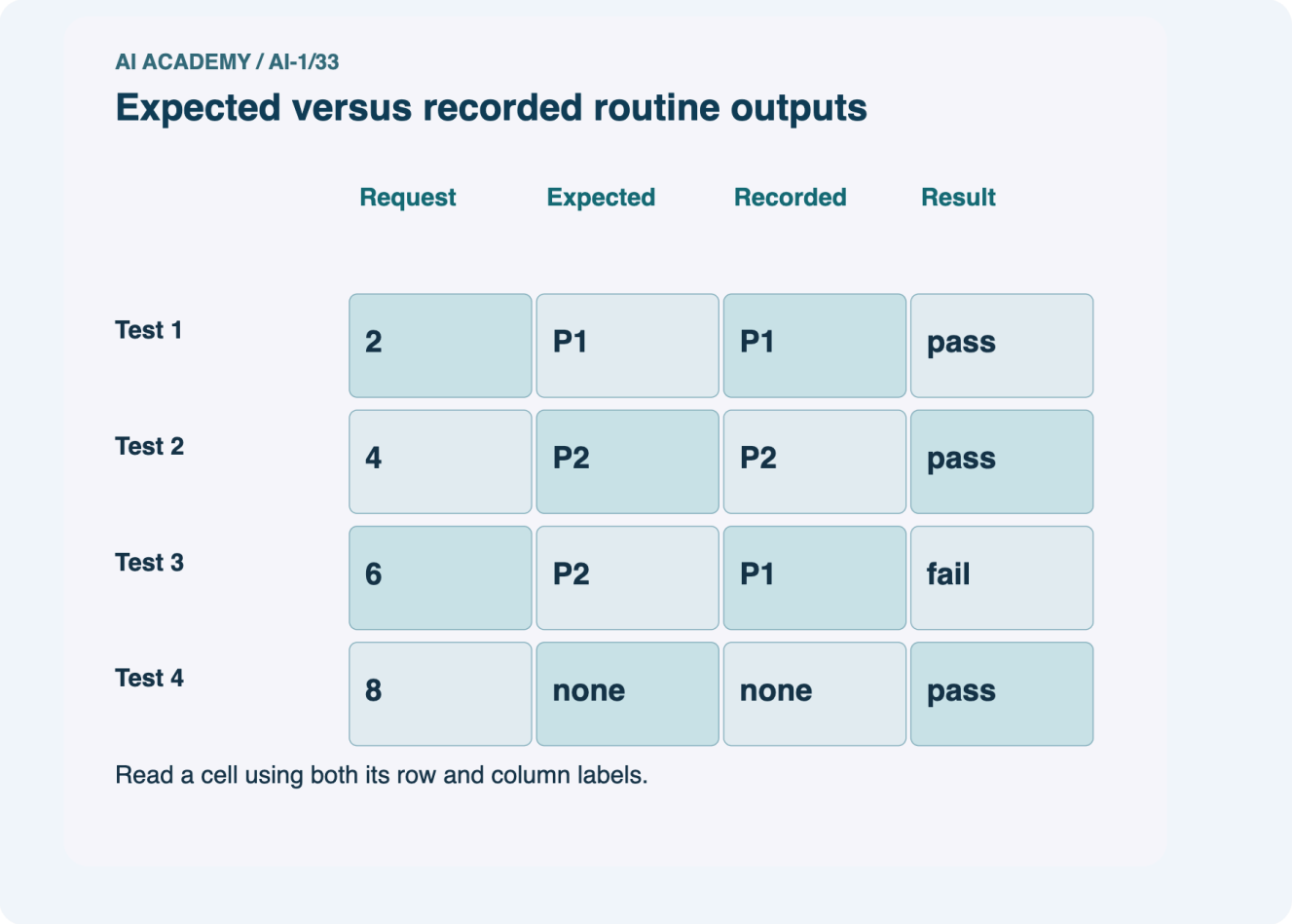
Follow the labelled picture with the matching explanation. Open the vector diagram

Write expectations: For 2,4,6,8 expect P1,P2,P2,none. → Compare **output**s: Runs at 2,4 and 8 agree; run at 6 disagrees. → Trace failure: P1 holds only 3, so it cannot satisfy request 6. → Report: Three of four routine cases pass; retain the failure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Expected versus recorded routine outputs



Follow the labelled picture with the matching explanation. Open the vector diagram

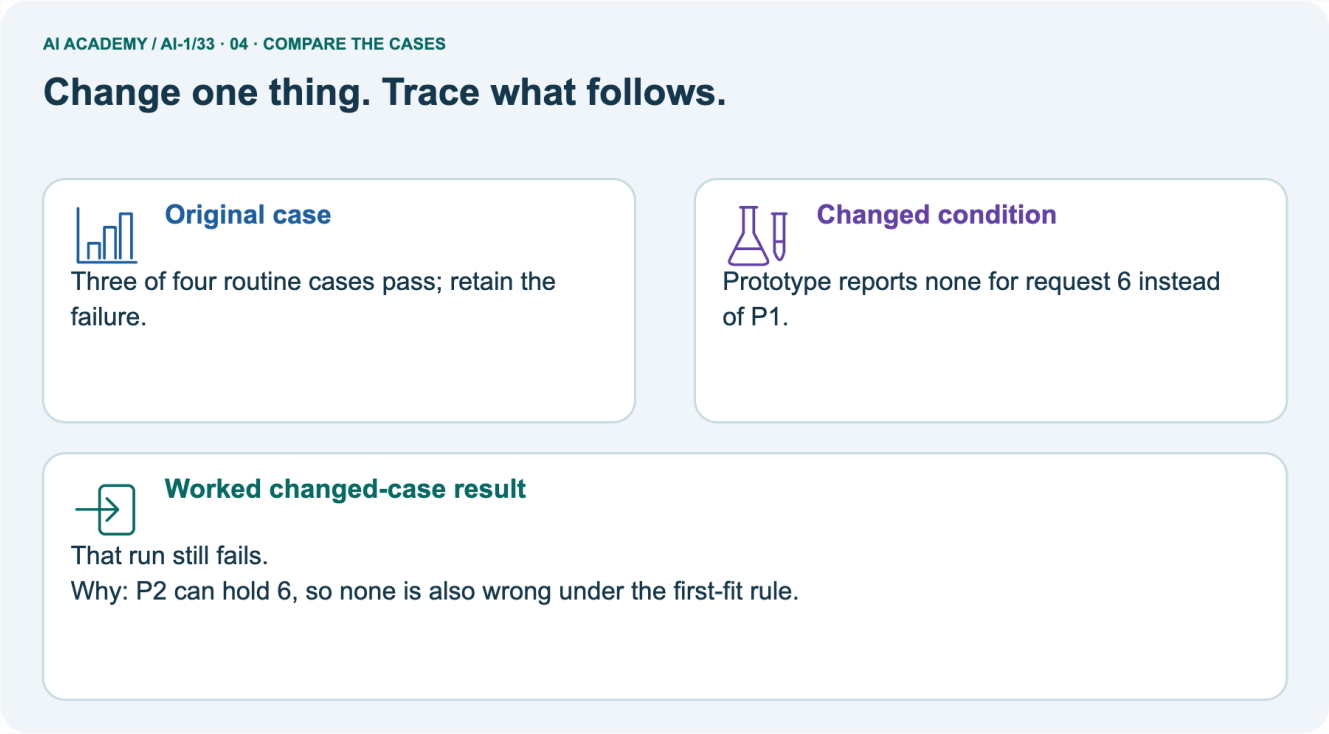
Expected versus recorded routine **output**s. Test 1: Request = 2, Expected = P1, Recorded = P1, Result = pass. Test 2: Request = 4, Expected = P2, Recorded = P2, Result = pass.

Test 3: Request = 6, Expected = P2, Recorded = P1, Result = fail. Test 4: Request = 8, Expected = none, Recorded = none, Result = pass.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

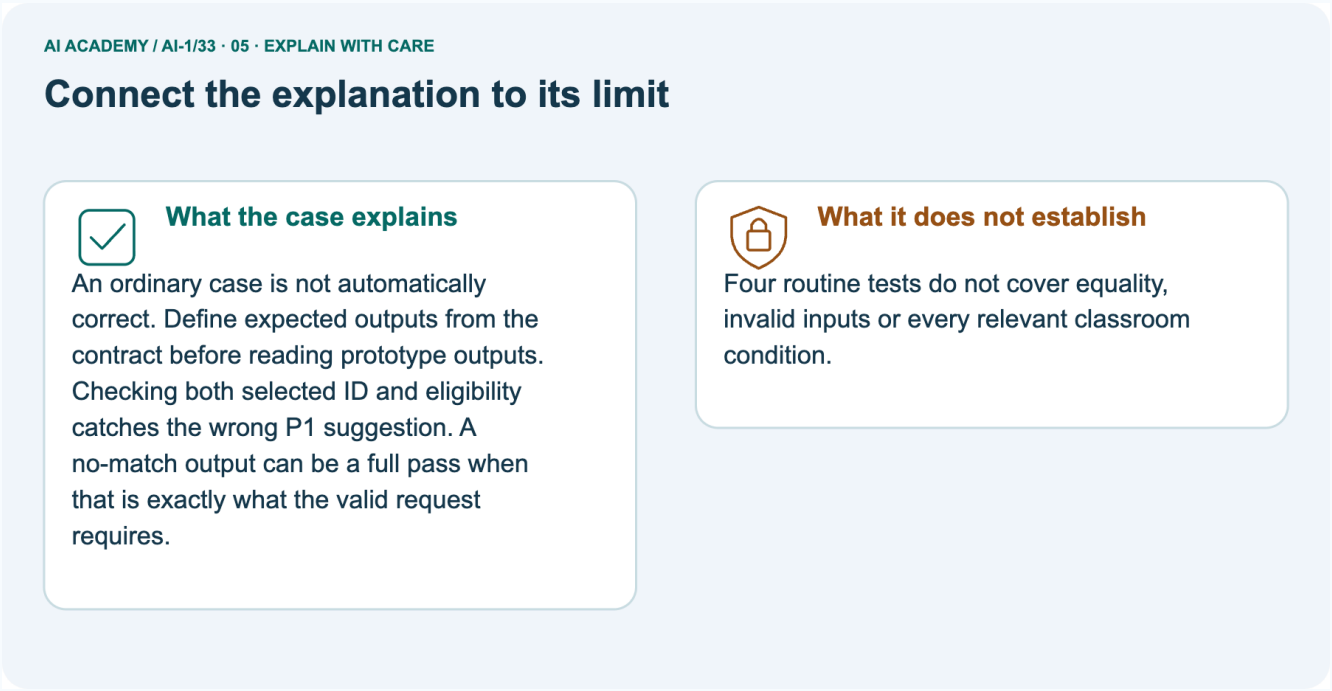
Original case: Three of four routine cases pass; retain the failure. Changed condition: Prototype reports none for request 6 instead of P1. **Expected result**: That run still fails.

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

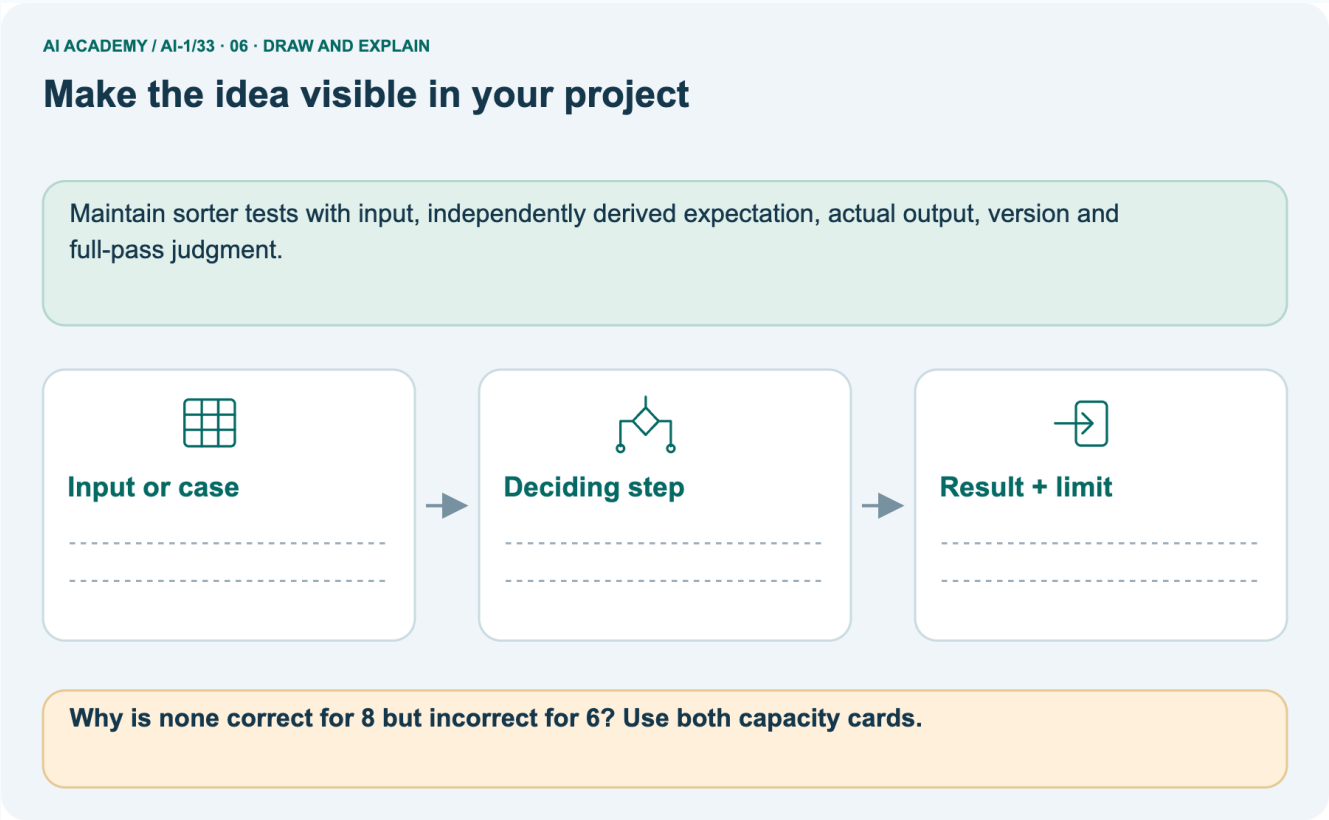
An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires. Limit: Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment. Draw your **input**, deciding step, result and limit.

Supported question: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Test Normal Cases: the case

02. Follow the deciding steps

03. Expected versus recorded routine outputs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.